

## STORA SPANSBERGET, Sweden

WMO: 024680

Lat: 60.382N

Lon: 15.137E

Elev: 493

StdP: 95.54

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.7      | -14.0 | -19.0                         | 0.7 | -15.8 | -16.6 | 0.9 | -12.5 | 8.3                   | -3.5 | 7.4 | -3.1 | 1.6                   | 20   | 0.638 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.2                    | 24.5            | 16.2 | 22.6 | 15.4 | 20.8 | 14.6 | 17.8                | 22.0 | 16.7 | 20.5 | 15.7 | 19.1 | 2.4                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.3                            | 12.3 | 18.6 | 15.2 | 11.5 | 17.9 | 14.3 | 10.8 | 16.9 | 52.1          | 22.1 | 48.6 | 20.4 | 45.5 | 19.1 | 21.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 7.2               | 6.3  | 5.5 | -18.1                      | 26.9 | 2.9                | 1.4 | -20.2                                              | 27.9 | -21.9      | 28.7 | -23.6      | 29.5 | -25.7      | 30.5 |  |  |
|                   |      |     | DB                         | WB   | DB                 | WB  | DB                                                 | WB   | DB         | WB   | DB         | WB   | DB         | WB   |  |  |
|                   |      |     | -18.4                      | 19.0 | 2.8                | 1.5 | -20.4                                              | 20.1 | -22.0      | 21.0 | -23.6      | 21.8 | -25.7      | 22.9 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.3    | -5.2  | -5.0  | -1.9  | 3.4   | 8.7   | 12.8  | 15.4  | 13.8  | 9.3   | 3.8   | -0.7  | -3.6 |
|                                                                                  | DBStd       | 8.21   | 4.29  | 4.40  | 4.33  | 3.81  | 4.07  | 3.54  | 3.31  | 2.85  | 2.94  | 3.52  | 3.89  | 4.44 |
|                                                                                  | HDD10.0     | 2533   | 472   | 420   | 368   | 202   | 74    | 11    | 1     | 3     | 46    | 195   | 321   | 421  |
|                                                                                  | HDD18.3     | 5145   | 731   | 653   | 627   | 449   | 298   | 168   | 104   | 143   | 269   | 452   | 571   | 679  |
|                                                                                  | CDD10.0     | 449    | 0     | 0     | 0     | 3     | 34    | 97    | 167   | 121   | 26    | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 19     | 0     | 0     | 0     | 0     | 0     | 4     | 13    | 3     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 105    | 0     | 0     | 0     | 0     | 2     | 28    | 66    | 9     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 8      | 0     | 0     | 0     | 0     | 0     | 1     | 7     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.8    | 2.6   | 2.6   | 3.1   | 2.9   | 2.8   | 2.6   | 2.5   | 2.4   | 2.7   | 3.1   | 3.1   | 2.7  |
| Precipitation                                                                    | PrecAvg     | 675    | 43    | 33    | 33    | 41    | 52    | 73    | 86    | 84    | 59    | 64    | 57    | 50   |
|                                                                                  | PrecMax     | 874    | 92    | 62    | 67    | 103   | 133   | 144   | 230   | 161   | 128   | 163   | 148   | 93   |
|                                                                                  | PrecMin     | 546    | 7     | 6     | 2     | 6     | 10    | 16    | 24    | 15    | 10    | 19    | 21    | 7    |
|                                                                                  | PrecStd     | 87     | 21    | 15    | 18    | 26    | 27    | 31    | 46    | 35    | 30    | 34    | 30    | 23   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 4.1   | 4.7   | 10.9  | 17.1  | 23.2  | 25.9  | 27.6  | 24.5  | 18.9  | 13.5  | 8.8   | 6.5  |
|                                                                                  |             | MCWB   | 2.9   | 1.9   | 5.3   | 9.6   | 13.2  | 15.7  | 17.8  | 16.6  | 14.0  | 11.5  | 8.1   | 6.2  |
|                                                                                  | 2%          | DB     | 2.4   | 3.3   | 7.7   | 14.2  | 20.5  | 23.9  | 25.1  | 22.2  | 16.7  | 11.1  | 7.2   | 4.5  |
|                                                                                  |             | MCWB   | 1.8   | 1.7   | 3.6   | 7.9   | 12.2  | 15.0  | 17.1  | 16.0  | 12.8  | 10.0  | 6.5   | 4.0  |
|                                                                                  | 5%          | DB     | 1.3   | 2.0   | 5.6   | 11.7  | 18.1  | 21.6  | 23.2  | 20.4  | 15.1  | 9.8   | 5.8   | 3.2  |
|                                                                                  |             | MCWB   | 0.9   | 1.0   | 2.5   | 6.4   | 11.4  | 14.0  | 16.3  | 15.2  | 12.1  | 9.1   | 5.4   | 2.7  |
|                                                                                  | 10%         | DB     | 0.3   | 0.8   | 3.8   | 9.3   | 15.5  | 19.3  | 21.3  | 18.5  | 13.5  | 8.7   | 4.6   | 1.9  |
|                                                                                  |             | MCWB   | 0.0   | 0.1   | 1.3   | 4.9   | 9.9   | 12.8  | 15.5  | 14.5  | 11.3  | 8.0   | 4.3   | 1.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.3   | 3.5   | 5.7   | 10.4  | 15.1  | 17.6  | 19.6  | 18.5  | 15.1  | 11.9  | 8.1   | 6.3  |
|                                                                                  |             | MCDB   | 3.5   | 4.0   | 9.8   | 15.8  | 19.5  | 22.8  | 23.9  | 21.7  | 17.3  | 12.9  | 8.1   | 6.3  |
|                                                                                  | 2%          | WB     | 1.9   | 2.0   | 4.0   | 8.5   | 13.6  | 15.9  | 18.3  | 17.1  | 13.7  | 10.5  | 6.3   | 4.2  |
|                                                                                  |             | MCDB   | 2.2   | 2.5   | 6.8   | 13.3  | 18.8  | 20.9  | 22.8  | 20.1  | 15.7  | 10.8  | 6.6   | 4.4  |
|                                                                                  | 5%          | WB     | 1.0   | 1.1   | 2.9   | 6.9   | 12.1  | 14.7  | 17.3  | 16.0  | 12.7  | 9.3   | 5.2   | 2.8  |
|                                                                                  |             | MCDB   | 1.4   | 1.8   | 5.1   | 10.7  | 16.5  | 19.8  | 21.7  | 18.7  | 14.4  | 9.6   | 5.4   | 3.0  |
|                                                                                  | 10%         | WB     | 0.0   | 0.1   | 1.8   | 5.5   | 10.8  | 13.6  | 16.3  | 15.1  | 11.7  | 8.2   | 4.0   | 1.4  |
|                                                                                  |             | MCDB   | 0.3   | 0.7   | 3.4   | 8.4   | 15.2  | 18.1  | 20.3  | 17.8  | 13.1  | 8.7   | 4.3   | 1.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.8   | 4.2   | 5.6   | 7.3   | 8.6   | 8.8   | 8.2   | 7.1   | 6.0   | 4.4   | 3.4   | 3.6  |
|                                                                                  |             | MCDBR  | 4.4   | 4.6   | 7.6   | 10.4  | 11.9  | 11.8  | 10.8  | 9.4   | 7.8   | 5.3   | 4.7   | 4.7  |
|                                                                                  |             | MCWBR  | 4.4   | 3.7   | 4.8   | 5.8   | 6.4   | 5.8   | 5.0   | 4.8   | 4.6   | 4.3   | 4.8   | 4.7  |
|                                                                                  | 5% WB       | MCDBR  | 4.2   | 4.3   | 7.0   | 9.8   | 10.5  | 10.8  | 9.8   | 7.9   | 6.9   | 4.9   | 4.7   | 4.8  |
|                                                                                  |             | MCWBR  | 4.4   | 3.8   | 4.6   | 5.7   | 6.1   | 5.8   | 5.1   | 4.8   | 4.6   | 4.4   | 4.7   | 4.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.238  | 0.282 | 0.306 | 0.340 | 0.340 | 0.334 | 0.353 | 0.346 | 0.315 | 0.302 | 0.259 | 0.212 |      |
|                                                                                  | taud        | 2.214  | 2.336 | 2.405 | 2.414 | 2.461 | 2.499 | 2.457 | 2.470 | 2.548 | 2.497 | 2.280 | 2.096 |      |
|                                                                                  | Ebn at Noon | 552    | 730   | 832   | 862   | 887   | 898   | 869   | 847   | 821   | 707   | 516   | 412   |      |
|                                                                                  | Edh at Noon | 38     | 62    | 80    | 95    | 98    | 96    | 98    | 90    | 70    | 53    | 35    | 26    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.28   | 0.93  | 2.46  | 3.88  | 4.76  | 5.41  | 4.94  | 3.84  | 2.53  | 1.14  | 0.35  | 0.17  |      |
|                                                                                  | RadStd      | 0.05   | 0.17  | 0.29  | 0.45  | 0.48  | 0.36  | 0.60  | 0.42  | 0.35  | 0.21  | 0.09  | 0.02  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (48 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page